HAWASSA UNIVERSITY

BENSA DAYE CAMPUS

DEPARTMENT OF COMPUTER SCIENCE

WEB PROGRAMMING (InTe3113) LABORATORY MANUAL

PROGRAM: B.SC. DEGREE IN COMPUTER SCIENCE

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PART ONE: STRUCTURING DOCUMENTS FOR THE WEB

LABORATORY I: Basics of HTML Structure and Text Formatting

Objectives:

- Acquire skills of the basic structure of hypertext Markup Language
- Implement the basic structure tags of web page development
- Commenting the tags
- Use HTML basic tags

HTML Basic Structure

HTML stands for Hyper Text Markup Language, which is the most widely used language on Web to develop web pages.

- ✓ Hypertext –refers to the way in which Web pages (HTML documents) are linked together. Thus, the link available on a web page are called Hypertext
- ✓ Markup – using HTML to simply markup a text document with tags that tell a web browser how to structure it to display

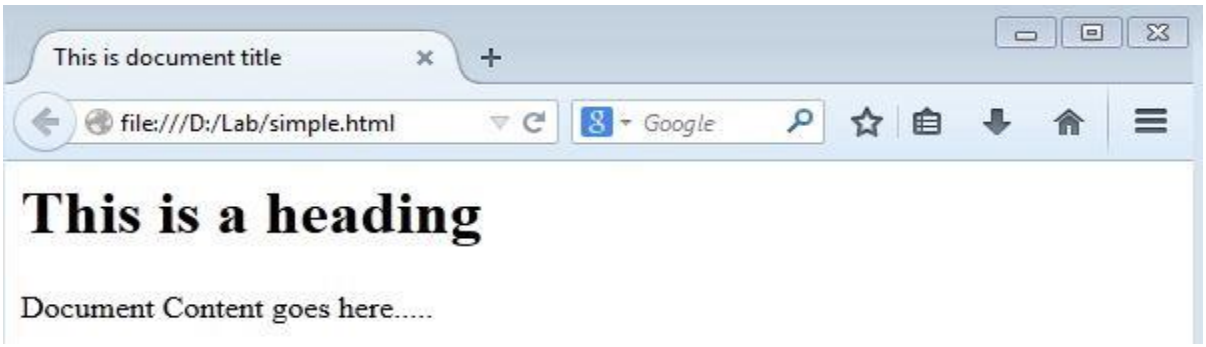
To create a simple webpage, follow the following steps

Step 1: open notepad /notepad++ (windows-> all programs->accessories->notepad) or Visual Studio Code

Step 2: write the codes shown below on the text editor

```
<! DOCTYPE html       >
<html>
<head>
<title>This is document title</title>
</head>
<body>
<h1> This is a heading </h1>
<p> Document Content goes here…..</p>
</body>
</html>
```

Step 3: save the file with the extension of file name .html example simple.html Step 4: Open the file with the web browser to produce the following output.



The above example of HTML document uses the following tags:

<! DOCTYPE ….> this tag defines the document type and HTML version

<html> this tag encloses the complete HTML document and mainly comprises of document header which is represented by <head>…</head>, document title which is represented by <title>…</title> and document body which is represented by <body>…</body> tags.

<head> this tag represents the documents header which can keep other HTML tags like <title>, <link> etc. It may contain elements that are designed for search engines to process or elements that change the overall appearance of the webpage. But elements in the header do not display directly as normal webpage content.

<title> the title <title> tag is used inside the <head> tag to mention the document title

<body> this tag represents the document's body which put the text and elements to be displayed in the main browser windows and also keeps other HTML tags like <h1>, , <p> etc

<h1> this tag represents the heading

<p> this tag represents a paragraph

HTML Basic Tags

Heading Tags

Any document starts with a heading. HTML have six levels of headings, which use the elements <h1>, <h2>, <h3>, <h4>, <h5> and <h6>. While displaying any heading, browser adds one line before and one line after that heading.

Example: Heading tags

```
<!DOCTYPE html>
< html>
< head>
<title>Heading Example</title>
< /head>
< body>
<h1>This is heading 1</h1>

<h2>This is heading 2</h2>
< h3>This is heading 3</h3>
< h4>This is heading 4</h4>
<h5>This is heading 5</h5>

<h6>This is heading 6</h6>
< /body>
</html>
```

The above program will produce the following result

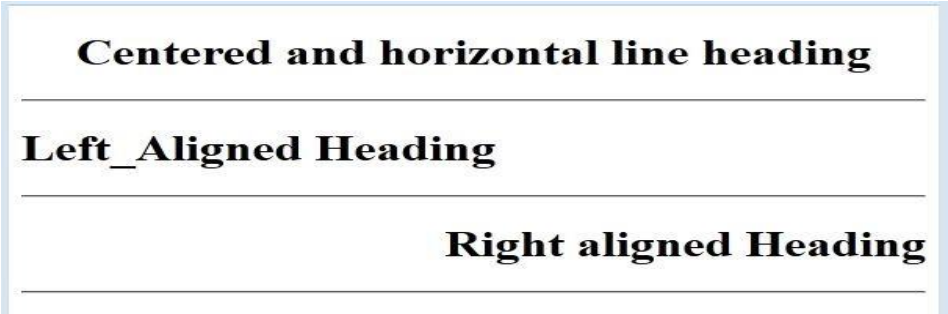
Align headings and Horizontal Line

It is possible to align headings in left, right and center. The HTML code shown below demonstrates the align headings in left, right, center and horizontal line. The align attribute uses left, right and center value to align text headings in their respective sides of the browser. And the tag <hr /> is used to create horizontal line.

Example: alignment and horizontal line

```
<!DOCTYPE html>
<html>
<head>
<title>  align and horizontal line Example</title>
</head>
<body>
<h1 align="center">Centered and horizontal line heading </h1><hr />
<h1 align="left"> Left Aligned Heading</h1><hr />
<h1 align="right"> Right aligned Heading</h1><hr />
</body>
</html>
```

This will produce the following result



N: B. The content of the paragraph can be centered using …. tag.

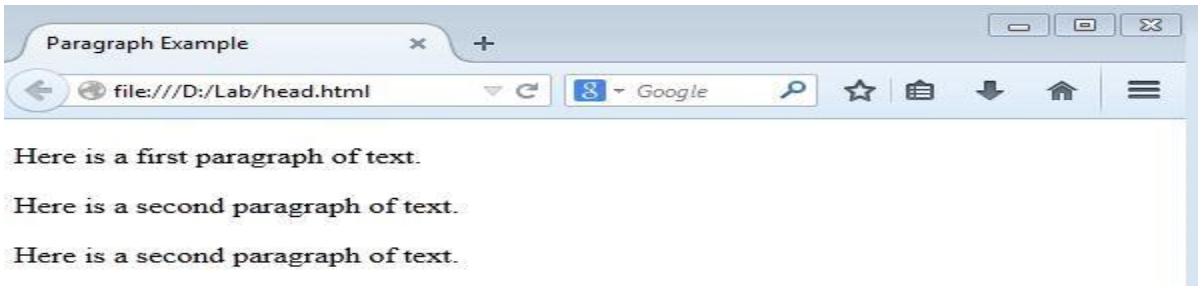
Paragraph Tag

The tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening and closing tags as shown below in the example: when a browser displays a paragraph, it usually inserts a new line before the next paragraph and adds a little bit of extra vertical space

Example: Paragraphing

```
<! DOCTYPE html>
<html>
<head>
<title> Paragraph Example</title>
</head>
<body>
<p> Here is a first paragraph of text. </p>
<p> Here is a second paragraph of text. </p>
<p> Here is a second paragraph of text. </p>
</body>
</html>
```

This will produce the following result



Line Break Tag

The <br /> element is used when opening and closing of tags is not needed and to start from the next line. There should be a space between the character br and the forward slash because older browsers will have trouble rendering the line break.

This will produce the following result

```
Good morning
This is the sample program of line break
Is that working
thanks
have nice time
```

Preserve Formatting

Any text between the opening <pre> and the closing </pre> tag will preserve the formatting of the source document. This tag keeps the text to follow the exact formate of how it is written in the HTML document.

Example : preserve formatting

```
<!DOCTYPE html>
<html>
<head>
<title> Preserve Formatting Example</title>
</head>
<body>
<pre>
function preservingFunction(strText){
alert (strText) }
</pre>
</body>
</html>
```

This will produce the following result?

Example: Line break

```
<!DOCTYPE html>
<html>
<head>
<title> Line Break Example</title>
</head>
<body>
<p> Good morning <br />  This is the sample program of line break <br />
Is that working <br />  thanks <br /> Have nice time<br /></p>
</body>
</html>
```

```
function preservingFunction(strText){
alert (strText)
}
```

Comments – comments use the following syntax

<!--comment goes here -->
Anything after *<!--until the closing -->* will not be displayed. It can still be seen in the source code for the document, and any other notes to anyone looking at the code. It is good practice to comment your code, especially in complex documents, to indicate sections of a document, and any other notes to anyone looking at the code.

Non-breaking Spaces

Use a non breaking space entity * * to give space between words. For example when coding the "the Student at BDU" in a paragraph, we should use the similar to the following code

```
<!DOCTYPE html>
<html>
<head>
<title> Non breaking Space Example</title>
</head>
<body>
<p>
this program is developed by "the Student at BDU."
</p>
</body>
</html>
```

This will produce the following result

this program is developed by "the Student at BDU."

Presentational Elements

We are familiar with the ability to make text bold, Italic, or underline in Microsoft word processor. In HTML, the following presentational elements are used:

The Element

Anything that appears in a element is displayed in bold, like the word "**bold"** here.

The syntax is bold

The content of a element will be displayed in the same way as the content of the element. It is used to indicate that its content have strong emphasis.

The syntax is content

The Element

the content of an element is displayed in italicized text, like the word "*italic*" here.

The syntax is italic.

The content of an element will be displayed in the same way as the contents of the element. It is used to indicate that its contents have emphasis.

The <u> Element

The content of a <u> element is underlined with a simple line. The syntax is <u>Underlined</U> to produce results like <u>Underlined</u>.

The <s> and <strike> Element

<s> is just the abbreviated form of <strike> which is used to display texts with strikethrough.

Strikethrough is a thin line through the text.

The <tt> Element

The content of a <tt> element is written in monospaced font like that of a teletype machine. It is used to make each character having the same size. The syntax is <tt>monospaced</tt>.

The <sup> Element

The content of a < sup > element is written in superscript; it is displayed half a character's height above the other characters and is also often slightly smaller than the text surrounding it. The <sup > element is especially helpful in adding exponential values to equations, and adding the *st*, *nd*, *rd*, and *th* suffixes to numbers such as dates.

Example : Written on the 31 < sup > st < /sup > February.

The <sub> Element

The content of a < sub > element is written in subscript; it is displayed half a character's height beneath the other characters and is also often slightly smaller than the text surrounding it.

Example*: the molecular formula of water is H < sub > 2 < /sub >O.*

The <big> Element

The < big > element was introduced to make the contents of this element one font size larger than the rest of the text surrounding it.

Example: *The following word should be < big > bigger < /big > than those around it.*

The <small> Element

The < small > element is the opposite of the < big > element, and its contents are displayed one font size smaller than the rest of the text surrounding it. If the font is already the smallest, it has no effect.

Example: *The following word should be < small > smaller < /small > than those around it.*

The <del> and <ins> Elements

Anything that appears within <ins>…</ins> element is displayed as inserted text where as the content that appears within <del>… </del> element is displayed as deleted text.

All the above presentational elements are coded as shown below

```
<!DOCTYPE html>
<html>
<head>
<title> Presentational Elements Example </title>
</head>
<body>
<p>I am a <b>Student</b> at BDU.</p>
<p> I am doing <i>Internet Engineering</i> program<br />
to <u>understand</u> <del>user-side</del> <ins>client side</ins> programming</p>
<p>If I make <s>mistake</s> in the source code, my Teacher give <tt>correction</tt></p>
<p>the date is 21<sup>st</sup> October</p>
<p> to show the subscript example like H<sub>2</sub>0 </p>
<p> whatever the text<big>bigger</big> or <small>Smaller</small>, It is readable</p>
</body>
</html>
```

This will produce the following result

I am a **Student** at BDU.

I am doing *Internet Engineering* program
to understand ~~user-side~~ client side programming

If I make ~~mistake~~ in the source code,my Teacher give correction

the date is 21st October

to show the subscript example like $H_2$0

whatever the textbigger or Smaller, It is readable

Grouping Content

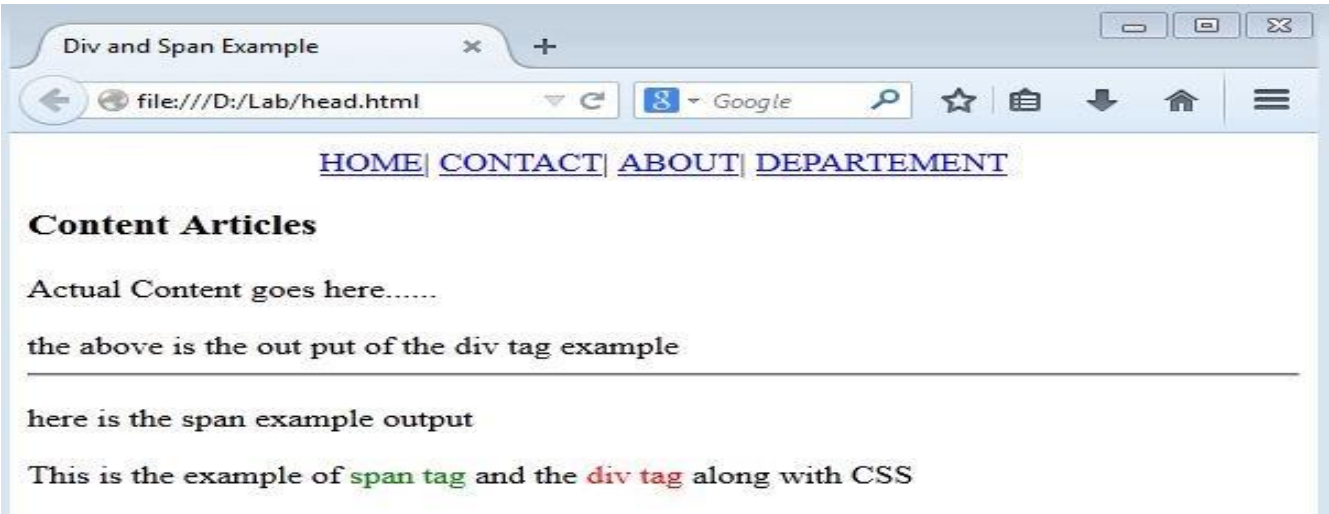
The and elements allow you to group together several elements to create sections or subsections of a page. For example: you might want to put all of the footnotes on a page within a element to indicate that all of the elements within that element relate to the footnotes.

The element, on the other hand, can be used to group inline elements only. So, if you have a part of a sentence or paragraph which you want to group together, you could use the element as follow

Example: Grouping the content

```
<! DOCTYPE html>
<html>
<head>
<title> Div and Span Example </title>
</head>
<body>
<div id="menu" align="middle">
<a href="/index.html">HOME</a>|
<a href="/about/contact_us.html">CONTACT</a>|
<a href="/about/index.html">ABOUT</A>|
<a href="/about/dept_info.html">DEPARTEMENT</a>
</div>
<div id="content" align="left" bgcolor="white">
<h3>Content Articles</h3>
<P> Actual Content goes here.....</p>
the above is the output of the div tag example<hr />
</div>
<P> here is the span example output</p>
<p>This is the example of <span style="color:green">span tag</span> and the <span
style="color:red"> div tag</span> along with CSS</p>
</body>
</html>
```

This will produce the following result



Phrase elements

Some of the elements in the phrase elements are working similar to the presentational elements. For example the words written in an ** element will look like the words in an *<i>* element but the ** element is supposed to indicate the addition of emphasis.

and for emphasis

The content of an and elements are intended to be a point of emphasis and strong emphasis respectively in the part of document. Browsers display the strong emphasis in a bold font whereas the emphasis in an italic font.

Example: the emphasis should be given if texts are written in bold and italic font

<cite>, and <q> for quotations and citation

The <q> element is used to add a quote within a sentence. The text enclosed in a <q>…</q> elements will be embedded in double quotes.

For example: As Albert Einstein said, <q> Learn from yesterday, live for today, hope for tomorrow</q>.

The <cite> element is used if we are quoting a text. The source should be placed between an opening <cite> tag and closing </cite> tag. The content of the source is displayed in an italicized font.
Example: <P> these tags are taken from <cite>Internet and world wide web</cite> book

Marked Text

Any text that appear within the <mark>…</mark> element is marked with yellow ink.

Text Abbreviation

To abbreviate a text, it should be placed inside the opening <abbr> and closing </abbr> tags. The title attribute should contain the full description.

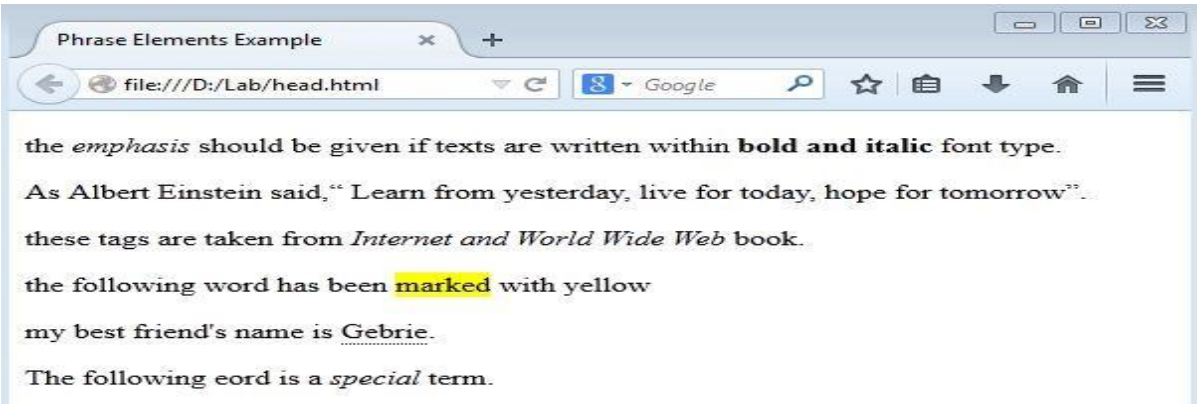
Special Terms

The <dfn> ….</dfn> element or HTML definition element allows you to specify that you are introducing a special term. Its usage is similar to italic words in the midst of a paragraph.

Example: Phrase elements

```
<!DOCTYPE html>
<html> <head> <title> Phrase Elements Example </title></head>
 <body>
<p> the <em> emphasis </em> should be given if texts are written within <strong> bold and italic</strong> font type. </p>
<p> As Albert Einstein said, <q> Learn from yesterday, live for today, hope for tomorrow</q>.</p>
<p> these tags are taken from <cite> Internet and World Wide Web</cite> book. </p>
<p>the following word has been <mark> marked</mark> with yellow</p>
<p> my best friend's name is <abbr title="Geberemedehin">Gebrie</abbr>.</p>
<p> The following eord is a <dfn> special</dfn> term.</p>
</body>
</html>
```

This will produce the following result



Exercise

1. Open the text editor and write HTML basic structure that contains title "Hawassa university", heading "Bensa Daye Campus" and the body the following text "Bensa Daye Campus is One of the campus in Hawassa university that includes eight department"

Then save it as example.html and run it to see the result.

2. Write the HTML program that include heading 1 “Ethiopia” within which paragraphs “Ethiopia is the east Africa country. Ethiopia has many tourist attracting places like lalibla, axume, lake tana highland monasteries etc” and heading 2 “ Regions” within which paragraphs “ there are two cities and nine regions.”

3. Write HTML program to get the following output

HOME | SERVICES| LOCATION| STAFFS| HELP

Introduction

Our hotal is found in the **amhara region** at ~~Gondar~~ Bahir Dar
town which is the capital of the region. the hotel gives *bathing,bed rooms, restruant, cafe services*.
if you need to visite monastries on the highland of tana,the hotel provides super^{star} boats.
Our hotel name is RSSH.
come$_{to}$visit$_{us}$!!!.
big thanks!!
21st January

LABORATORY II: List and Link

Objectives:

- Use unordered list tag in HTML script program
- Use ordered list tag in HTML script program
- Write link tags in the HTML script programs to refer to another page

Lists

Unordered Lists

The <ul> tag stands for unordered list which is used to make a list of bullet points. Each bullet point or line you want to write should then be contained between opening <li> tags and closing </li> tags. li- stands for list item. The type of the bullet can be square, disc, circle and so on. Show the following example:

Example: Unordered lists

```
<!DOCTYPE html>
<html>
<head>
<title> Unordered Lists Example </title>
</head>
<body>
<p> the following is Unordered list using square bullet</p>
<ul type="square">
        <li> Hawassa University</li >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
        <li> Gondar University</li>
        <li> BDU University</li    >
</ul>
<p> the following is the Unordered list using the default bullet or disc bullet</p>
 <ul type="disc">
        <li> Hawassa University</li >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
        <li> Gondar University</li>
        <li> BDU University</li    >
</ul>
<p> the following is the Unordered list using the circle bullet</p>
 <ul type="circle">
        L> Hawassa University</li   >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
        <li> Gondar University</li>
        <li> Awassa University</li>
</ul></body></html>
```

Ordered Lists

In an order list, rather than prefixing each point with a bullet point, we can use either numbers (1,2,3), letters(A,B,C), or roman numerals(i,ii,iii) to prefix the list item.

An order list is contained inside the *<ol>* element. Each item in the list should then be nested inside the *<ol>* element and contained between opening *<li>* and closing *</li>* tags

If you would rather have letters or roman numerals than Arabic numbers, we can use the following attribute type on the <ol> element.

Value for type attribute	**Description**	**Example**
1	Arabic numerals(the default)	1, 2, 3, 4, 5
A	Capital letters	A, B, C, D, E
a	Small letters	a, b, c, d, e
I	Large roman numerals	I, II, III, IV, V
i	Small roman numerals	i, ii, iii, iv, v

The figure shown below is the output of the ordered lists of the HTML program shown in the example below with the Arabic numerals, capital letters, small letters, large roman numerals and small roman numerals.

the following is ordered list using arabic numerals (the default)

1. Bahir dar University
2. Addis Abeba University
3. Arba Minch University

the following is the ordered list using the small roman numerals

i. Bahir dar University
ii. Addis Abeba University
iii. Arba Minch University

the following is the ordered list using Large roman numerals

I. Bahir dar University
II. Addis Abeba University
III. Arba Minch University

the following is the Unordered list using capital letters

A. Bahir dar University
B. Addis Abeba University
C. Arba Minch University

the following is the Unordered list using small letters

a. Bahir dar University
b. Addis Abeba University
c. Arba Minch University

Example: Ordered List

```
<!DOCTYPE html>
<html>
<head>
<title> ordered Lists Example </title>
</head>
<body>
<p> the following is ordered list using arabic numerals (the default)</p>
<ol type="1">
        <li> Hawassa University</li >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
</ol>
<p> the following is the ordered list using the small roman numerals </p>
 <ol type="i">
        <li>Hawassa University</li  >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
</ol>
<p> the following is the ordered list using Large roman numerals</p>
 <ol type="I">
        <li> Hawassa University</li>
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
</ol>
<p> the following is the unordered list using capital letters</p>
 <ol type="A">
        <li> Hawassa University</li >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
</ol>
<p> the following is the unordered list using small letters</p>
 <ol type="a">
        <li> Hawassa University</li >
        <li> Addis Abeba University</li>
        <li> Arba Minch University</li>
</ol>
</body>
</html>
```

Definition Lists

The definition list is a special kind of list for providing terms followed by a short text definition or description for them. Definition lists are contained inside the < dl > element. The < dl > element then contains alternating < dt > and < dd > elements. The content of the < dt > element is the term you will be defining. The < dd > element contains the definition of the previous < dt > element.

Nested Lists

Lists can be nested inside other list. The items in the list may have its own list items so that this list items can be nested inside the outer list item as shown below.

Example: definition and nested lists

```
<!DOCTYPE html>
<html>
<head><title> Definition and Nested Lists Example </title></head>
<body>
<p> The following is example of definition list</p>
<dl>
        <dt>Unordered List</dt>
        <dd> A list of bullet points.</dd>
        <dt>An Ordered list</dt>
        <dd> An ordered list of points, such as a numbered set of steps.</dd>
        <dt>definition List</dt>
        <dd>A list of terms and definitions.</dd>
</dl>
<p> the following is example of nested list</p>
<ol type="I">
        <li>Addis Ababa University</li>
        <li> Hawassa University</li  >
                <ol type="i">
                        <li>Bensa Daye Campus</li>
                        <li>Computer Science Dep't</li>
                        <li> Plant Science dep't</li        >
                        </ol>
        <li>Arba Minch University</li>
</ol>
</body></html>
```

This will produce the following result : _____

The start Attribute

Start attribute is sometimes used for ordered list to specify the starting point of number we need.

The syntax of the start attribute for different numbering system will be the following:

<ol type="1" start ="4"> -numerals start with 4.
<ol type="I" start ="4"> -numerals start with IV.

<ol type="i" start ="4"> -numerals start with iv.

<ol type="a" start ="4"> -Letters start with d.

<ol type="A" start ="4"> -Letters start with D.

Basic Links

A link is specified using HTML tag <a>. This tag is called anchor tag and anything between the opening <a> tag and the closing </a> tag becomes part of the link that user can click in a browser.

The syntax of the anchor is *<a href="Document URL" . . . attributes-list>Link Text</a>* href- is an attribute that references the link web page URL and title attribute is also used in the <a> tag to display in a tooltip when we hover over the link and also the target attribute is used to open the link page either on new browser or on the same browser windows.

Syntax: * <a href =http://www.Google.com/ title="Search the web with Google" target="_blank">Google</a> is a very popular search engine. *

Example: link and definition the results will be

```
<!   DOCTYPE html>
<  Html>
<  Head>
<  title> Definition and Nested Lists
Example </title>
<  /head>
<  body      >
<  p> <a href="http://www.bdu.edu.et"
title="Click on the link to open the
website">
Hawassa University</a> is the website of
Hawassa University</p>
<  /body       >
<  /html       >
```

Base Path tag

To link HTML documents related to the same website, it is not required to give a complete URL for every link rather use <base> tag in the HTML document header- used to give a base path for all the links.

```
<!DOCTYPE html>
<html>
<head>
<title>Hyperlink Example</title>
<base href="http://www.hu.edu.et  /">
</head>
<body>
<p>Click following link</p>
<a href="/hu  /index.html" target="_blank">Hawassa University  </a>
</body>
</html>
```

Here, <a href="/hu/index.html" is being considered as <a href=http://www.hu.edu.et/hu/index.html .

A page Section Link

To create a link to a particular section of a given webpage by using name attribute, use the following two-step process

i. Create a link to the place where we want to reach within a webpage and name it using <a …> tag as follows:

   <h1>HTML Text Links <a name="top"></a></h1>

ii. To create a hyperlink to link the document and place where we want to reach:

< a href=”/hu/html_text_links.html#top”>Go to the Top</a>

Download Links

To create text link to make PDF, DOC or ZIP files downloadable, simply give complete URL of the downloadable files as follows:

<a href=http://www.hu.edu.et/page.pdf>Download PDF</a>

E-mail Addresses Link

Mailto links are links on web pages that instead of pointing to a web page URL or anchor, they point to an email address. Then when a reader clicks on the link, an email client opens and they can send a message to that email address. To create a link on your website that opens an email window, we simply use a mailto link. For example:

<a href=”mailto:webdesign@aboutguide.com”>Send email to the Web Design Expert</a>

LABORATORY III: Marquee and Multimedia

Objectives:

- Implement HTML marquee that rotate texts in specified direction
- Define different behavior of HTML marquee
- Adding image in the web page and using the image as the link
- Adding audio in the web page and using audio as the background of the page Adding video in the web page

HTML Marquees

An HTML marquee is a scrolling piece of text displayed either horizontally across or vertically down the page depending on the settings. This is created by using HTML <marquee> tag. The HTML marquee may not be supported by some browsers so that javascript or CSS can be used to create the same effect.

A simple syntax to use HTML <marquee> tag is:

<marquee attribute_name=”attribute_value” .... More attributes>

One or more lines or text message or image

</marquee>

The <marquee> Tag Attributes

Important attributes for the HTML <marquee> tag are:

- ✓ **width**-this specifies the width of the marquee
- ✓ **height** –this specifies the height of the marquee
- ✓ **direction**-this specifies the direction in which marquee should scroll like up, down, left or right.
- ✓ **behavior**-specifies the type of scrolling of the marquee like scroll, slide and alternate.
- ✓ **scrolldelay**-specifies how long to delay between each jump.
- ✓ **scrollamount**-specifies the speed of marquee text.
- ✓ **loop**-specifies how many times to loop. The default value is infinite (loops endless).
- ✓ **bgcolor**-specifies background color in terms of color name or color hex value.
- ✓ **hspace**-specifies horizontal space around the marquee.
- ✓ **Vspace**-specifies vertical space around the marquee.

Example 1: basics of marquee tag

```
<!DOCTYPE html>
<html>
<head>
<title> HTML marquee Tag </title>
</head>
<body>
<marquee>This is basic example of marquee tag</marquee>
</body>
</html>
```

Example 2: marquee with specified width

```
<!DOCTYPE html>
<html>
<head>
<title> HTML marquee Tag </title>
</head>
<body>
<marquee width="50%">This example will take only 50% width</marquee>
</body>
</html>
```

Example 3: the text scrolls from left to right with a speed of 10

```
<!DOCTYPE html>
<html>
<head>
<title> HTML marquee Tag </title>
</head>
<body>
<marquee width="50%" direction="right" scrollamount="10">
        this text will scroll from left to right and speed 10</marquee>
</body>
</html>
```

Adding Image

Images are added to a site using the <img> element, which consists at least two attributes: the *src* attribute to indicate the source of the image (It tells the browser where to find the image)- the value is the URL and an *alt* attribute to provide a description of the image.

Syntax to add photo*: <img src="photo.gif" alt="my photo" />*

The height and width attributes

The size of the image can be determine using height and width attribute –the value of these attributes can be specified in terms of either pixels or percentage of its actual size.

Syntax: *<img src="testimage.png" alt="Test Image" width="150" height="100" />*

Set Image Border and Alignment

By default image will have a border around it and align at the left side of the page, the thickness can be specified in terms of pixels using border attribute. A thickness of 0 means no border around the picture. Using align attribute, the image can be align at the center or right.

Syntax: *<img src="testimage.png" alt="Test Image" border="3" align="right" />*

Using Image as a Link

To use the image as a link, place the image between the opening <a> tag and closing </a> tag. The images are used to create graphical buttons or links to other pages.

Syntax: *<a href="../index.html" title="Click here to return to the Test page"><img src="testimage.png" width="150" height="100" border="3" alt="Test Image" /></a>* Produced Image example

```
<!DOCTYPE html>
<html>
<head>
<title> Image Example </title>
</head>
<body><p> this is image insertion and image link example</p>
<img src="testimage.png" width="150" height="100" border="3" alt="Test Image" />
<p> click on the logo to return the webpage of the university</p>
<a href="http://www.hu.edu.et" target="_blank" title="Clec  k here to open the home page"><img src="logo.png"
width="150" height="100" border="3" alt="logo Image" /></a>
</body>
</html>
```

result: _____

Adding Video and Audio

To add playable video and audio to the webpage for the user, the non-standard <embed> tag is required. Non-standard means the <embed> tag will not validate if we try to validate our webpage. Non standard tags are not listed in the HTML recommendation.

The <embed> tags includes *src* attribute, autostart, autoplay, width and height attributes. The autoplay and autostart attribute is used to play the audio and video automatically if they are set to

true otherwise set the value to false and the user should click on the play button to start playing the audio and video.

The syntax :*<embed src=”source URL” autostart=”false/true” autoplay=”false/true” width=” “ height=” “></embed>*

Adding Background music to the webpage

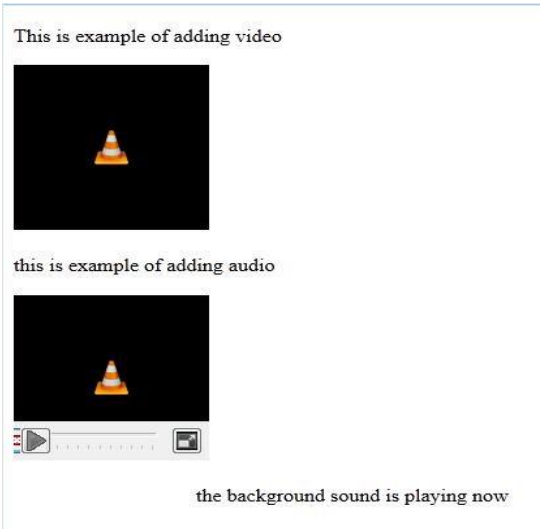
To add a background sound to the webpage, it can be used <embed>, <object>, and <bgsound> tags. Since <embed> tag supports most browser, we used this tag in our lesson. The <embed> tag includes src, autoplay, autostart and hidden attributes. The < embed> tag is used to embed music, videos and other kinds of multimedia in the webpage. The hidden attribute should be set to false to hide the sound clips control panel.

Syntax: *<embed src=”audio/video/image file” autostart=”true” autoplay=”true” hidden=”true”></embed>*

Example: Basic HTML multimedia example

```
<!DOCTYPE html>
<html>
<head>
<title> HTML Multimedia Tag </title>
</head>
<body>
        <p> this is example of adding video</p>
<embed src="video.mp4" type="video/mp4" autoplay="false" width="150" height="144"></embed>
        <p>this is example of adding audio</p>
<embed src="audio.mp3" type="audio/mp3" autoplay="false" width="150" height="144"></embed>
        <p><marquee scrollamount="5"> the background sound is playing now</marquee> </p>
<embed src="bgaudio.mp3" autoplay="true" hidden="true" loop="infinity"></embed>
</body>
</html>
```

The produced result is



Exercise:

1. Adding **"Fikir Lake"** image as the background image of your web page and use your photo as link image
2. Add your favorite music as the background audio of your web page
3. Add video music or other video files in to your web page

LABORATORY IV: HTML Tables

Objectives:

- Creating tables in HTML script using basic table elements
- Defining table background colors
- Define different table attributes
- Creating nested tables

Basic Table Elements and Attributes

To arrange data like text, image, links, other tables and so on into rows and columns of cells, the HTML tables are used. The tables are created using <table> tag in which the <tr> tag is used to create table rows, the <td> tag is used to create data cells and the <th> element is used to create table heading.

The table below contains three column and three rows of records created by the HTML script program shown below in the example

This is the HTML table with Heading

Name	Status	Salary
Solomon Tesfa	Lecturer	8400
Mengistu Belay	Ass.Lecturer	5200
yebeltal Tegegn	Lab Assistance	3000

Example: Creating simple table having heading, rows and columns

```
<!DOCTYPE html>
<html><head><title> HTML table example
</title></head><body>
<p> This is the HTML table with Heading</p>
<table border="1">
        <tr>
                <th> Name</th>
                <th> Status</th>
                <th>Salary</th>
        </tr>
        <tr>
                <td>Solomon Tesfa</td>
                <td>Lecturer</td>
                <td>8400</td>
        </tr>
        <tr>
                <td>Mengistu Belay</td>
                <td>Ass.Lecturer</td>
                <td>5200</td>
        </tr>
        <tr>
                <td>yebeltal Tegegn</td>
                <td>Lab Assistance</td>
                <td>3000</td>
        </tr>
</table></body></html>
```

Cellpadding and Cellspacing Attributes

To adjust the white space in the table cells cellpadding and cellspacing attributes are used. The cellspacing attribute defines the width of the border and cellpadding represents the distance between cell borders and the content within a cell.

Colspan and Rowspan Attributes

To merge two or more columns into a single column, colspan attribute is used and to merge two or more rows, rowspan is used.

Table Backgrounds

The table background is set using one of the following two ways:

- bgcolor attribute-using this attribute the background color for whole table or for one cell can be adjusted.

- Background attribute- using this attribute the background image for whole table or for one cell is adjusted.

Example: background color of the table

```
<!DOCTYPE html>
<html>
<head>
<title> HTML table example </title>
</head>
<body>
<p>HTML background, cellpadding,cellspacing,colspan and rowspan</p>
<table border="1" cellpadding="5" cellspacing="5" bordercolor="green" bgcolor="yellow">
<tr>
        <th>Column 1</th>
        <th>Column 2</th>
        <th>Column 3</th>
</tr>
<tr><td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr><td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr><td colspan="3">Row 3 cell 1</td></tr>
</table>
</body>
</html>
```

HTML background, cellpadding,cellspacing,colspan and rowspan

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 cell 1		

Table caption

The caption tag will serve as a title or explanation for the table and it shows up at the top of the table. In addition to the caption, tables can be divided into three portions: a header, a body, and footer. The three elements for separating the head, body, and foot of a table are:

- <thead> -to create a separate table header.
- <tbody> - to indicate the main body of the table.
- <tfoot> - to create a separate table footer.

A table may contain several <tbody> elements to indicate different pages or groups of data. But it is notable that <thead> and <tfoot> tags should appear before <tbody>

The height and width of the table can be adjusted in terms of pixels or percentage of available screen area.

The result produced for the example shown below is:

Table width/Height, Caption, header and footer are shown below

This is the caption

This is the head of the table			
row 1,column 1	row 1,column 2	row 1,column 3	row 1,column 4
row 2,column 1	row 2,column 2	row 2,column 3	row 2,column 4
This is the foot of the table			

Example: table caption

```
<!DOCTYPE html>
<html>
<head>
<title> HTML table example </title>
</head>
<body>
<p> Table width/Height, Caption, header and footer are shown below</p>
<table border="1" width="500" height="150">
<caption> This is the caption</caption>
<thead><tr><td colspan="4">This is the head of the table</td></tr></thead>
<tfoot><tr><td colspan="4">This is the foot of the table</td></tr></tfoot>
<tbody><tr>
                <td>row 1,column 1</td>
                <td>row 1,column 2</td>
                <td>row 1,column 3</td>
                <td>row 1,column 4</td>
        </tr>
        <tr>
                <td>row 2,column 1</td>
                <td>row 2,column 2</td>
                <td>row 2,column 3</td>
                <td>row 2,column 4</td>
        </tr>
</tbody>
</table>
</body>
</html>
```

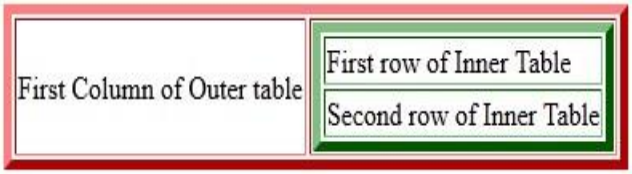
Nested Table

Nesting tables simply means making a table inside another table. Nesting table can lead to complex table layouts. Nested tables always need to be placed between <td>…</td> tags of the outer container table. The following HTML code creates a table with one row and two column and inside the second column again creates another table (nested table) with two rows.

Example: Nested table

Produced result will be:

```
<!DOCTYPE html>
<html>
<head>
<title> HTML Nested table example </title>
</head>
<body>
<table border="5" bordercolor="red">
<tr><td>First Column of Outer table</td>
        <td><table border="5"
bordercolor="green">
                <tr><td>First row of Inner
Table</td></tr>
                <tr><td>Second row of
Inner Table</td></tr>
                </table>
        </td>
</tr>
</table>
</body>
</html>
```



Exercise

1. Create the following table and add this table in to the web page and fill the row with records

Staff Members with their Salaries							
Name	**Months**						
	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.

2. Change the background color to green

LABORATORY V: HTML Forms

Objectives:

- Create single line text, multi-line text and password input text boxes
- Create check box and radio buttons in the web page
- Create submit and normal buttons in the web page using HTML scripts
- Implement file upload in the web page using HTML scripts
- Implement hidden file controls in HTML scripts

HTML Form Controls

HTML forms are required to collect some data from the site visitor. For example, during user registration to collect information such as name, email address, credit card, and so on. There are different types of form controls that are used to collect data using HTML form.

- ✓ Text input controls
- ✓ Checkboxes controls
- ✓ Radio box controls
- ✓ Select box controls
- ✓ File Select boxes
- ✓ Hidden Controls
- ✓ Clickable Buttons
- ✓ Submit and Reset Button

The HTML <form> tag is used to create an HTML form and it has the following sysntax:

<form action="Script URL" method="GET/POST">

Form elements like input, text area, and so on.

</form>

The most frequently used form attributes are:

- action – backend script ready to process passed data.
- method- is used to upload data (GET and POST methods)

- target- specifies the target windows or frame where the result of the script will be displayed.
- enctype- specifies how the browser encodes the data before it sends it to the server.

Text Input Controls

There are three types of text input used on forms:

- Single line text input controls – used to insert only one line user input like search box or name. it is created by <input> tag
- Password input controls- a single line text input which is masked the character as soon as a user enters it. It is created using HTML <input> tag.
- Multi-line text input controls- used to give details that may be longer than a single sentence. It is created by using HTML <textarea> tag.

Example: the following program contains single line text, password and multi-line text input controls.

```
<!DOCTYPE html>
<html>
<head>
<title> HTML Text Input Controls </title>
</head>
<body>
<form>
<p>Single line text input controls<p>
First name: <input type="text" name="first_name" /><br />
Last name:  <input type="text" name="last_name" /><br />
<p> Password input controls</p>
User ID: <input type="text" name="user_id" /><br />
Password: <input type="password" name="password" />
<p>Multiple-Line Text Input Controls</p>
Description:<br />
<textarea rows="5" cols="50" name="description">
Enter description here ...
</textarea>
</form>
</body>
</html>
```

The produced result will be:

Some of the attributes of the <input> tag are:

- ✓ type- indicates the type of input controls

- ✓ name-used to give a name to the control which is sent to the server to be recognized and get the value
- ✓ value-used to provide an initial value inside the control
- ✓ size-specify the width of the text-input control in terms of characters
- ✓ maxlength-allow to specify the maximum number of characters a user can enter into the text box
- ✓ rows-indicates the number of rows of text area box
- ✓ cols-indicates the number of columns of text area box

Selection Control

If multiple options are available to be selected by the user, depending on the selection requirement checkboxes, radio buttons and dropdown boxes can be used.

Checkboxes are used when more than one option is required to be selected. It is created using HTML <input> tag and the type attribute is set to checkbox.

Radio buttons are used when out of many options; just one option is required to be selected. It is created by using HTML <input> tag and type attribute is set to radio.

A select box, also called drop down box which provides option to list down various options in the form of drop down list, from where a user can select one or more options.

Some of the attributes of selection control are:

Type-indicates the type of input control. For checkbox the type set to checkbox and for radio buttons the type set to radio

Name- used to give to the control which is sent to the server to be recognized and get the value

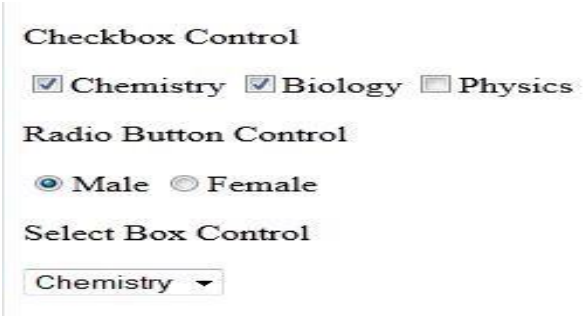
Value-the value that will be used if the checkbox is selected

Checked- set to checked if you want to select it by default

Example: the HTML code below shows the three selection controls

```
<!DOCTYPE html>
<html>
<head>
<title> HTML selection Controls </title>
</head>
<body>
<form>
<p>Checkbox Control<p>
<input type="checkbox" name="chemistry" value="chem">Chemistry
<input type="checkbox" name="Biology" value="bio">Biology
<input type="checkbox" name="Physics" value="phys">Physics<br />
<p>Radio Button Control</p>
<input type="radio" name="sex" value="M">Male
<input type="radio" name="sex" value="F">Female<br />
<p>Select Box Control</p>
<select name="dropdown">
        <option value="chem" selected>Chemistry</option>
        <option value="bio"> Biology</option>
        <option value="phys">Physics</option>
</select>
</form>
</body>
</html>
```

The produced result will be:



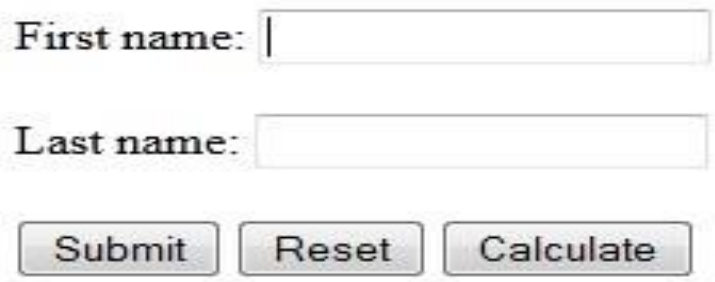
Creating Buttons

Buttons are most commonly used to submit a form, to clear or reset a form and even to trigger client –side scripts. The buttons are created using the HTML <input> tag and set the type attribute to submit, reset or button based on the required button type.

Example: the HTML code to show the type of buttons

```
<!DOCTYPE html>
<html>
<head>
<title> HTML button type example </title>
</head>
<body>
First name: <input type="text" name="first_name" /><br /><br />
Last name:  <input type="text" name="last_name" /><br /><br />
<input type="submit" name="btnSubmit" value="Submit" />
<input type="reset" value="Reset"/>
<input type="button" Value="Calculate" onclick="calculate()" />
</body>
</html>
```

The result produced for the three type of buttons are:



File Upload Box

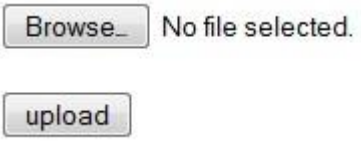
To allow users to upload a file to the website, it is needed to use a file upload box which is known as a file select box. It is created using the <input> element and type attribute is set to file.

Example: a file uploads box HTML source code

```
<!DOCTYPE html>
<html>
<head>
<title> HTML File upload example </title>
</head>
<body>
<form action=" method='POST' enctype='multipart/form-data'>
   <input type='file' name='userFile'><br>
   <input type='submit' name='upload_btn' value='upload'>
</form>
</body>
</html>
```

The result produced using internet explorer browser is

And the result produced using Firefox browser is



Hidden Form Controls

Hidden form controls are used to hide data inside the page which later on can be pushed to the server. This control hides inside the code and does not appear on the actual page. For example, following hidden form is being used to keep current page number. When a user will click next page then the value of hidden control will be sent to the web server and there it will decide which page has be displayed next based on the passed current page.

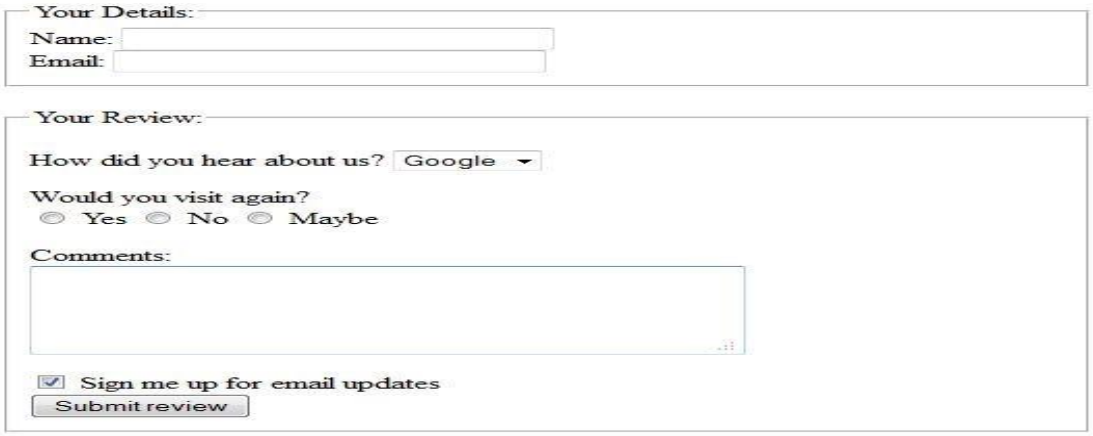
Example: the hidden form control HTML code

```
<!DOCTYPE html>
<html>
<head>
<title> HTML Hidden form control </title>
</head>
<body>
<form method="post" action="#URL#">
   <input type="hidden" value="08:03:31" name="startTime" />
   Username: <input type="text" name="userName" /> <br />
   Password: <input type="password" name="password" /><br /><br /><br />
   <input type="submit" value="Login" />
</form>
</body>
</html>
```

The result produced for the hidden form is

Exercise:

Write the HTML code that construct the following form



PART TWO: CASCADING STYLE SHEETS

The **C**ascading **S**tyle Sheets, referred to as CSS, is a simple design language intended to simplify the process of making web pages presentable. CSS handles the look and feel part of a web page. Using CSS, the color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colors are used, as well as a variety of other effects can be controlled.

The CSS syntax is made up of three parts: a selector, a property and a value.

Selector { property:value }

The selector is normally the HTML element or tag to be defined or changed the style. The property is the attribute to be changed and each property can take a value. The property and the value are separated by a colon and surrounded by curly braces.

There are four ways in which to apply CSS code to an HTML document. These are a) inline

b) embedded

c) External

d) Imported

LABORATORY VII: Inline and embedded Style Sheet

Objectives:

- Write style sheet rules along with the HTML element using style attribute of those elements
- Define inline style sheets to format the text and the web page
- Define the embedded style sheet to make the web page attractive
- Define the text, background and other element of the web page using inline and embedded style sheet

Inline Style Sheet

Style sheet rules can be applied directly to any HTML element using *style* attribute of the tag.

This should be done when there is a need to change the style of a particular HTML element. Rules define in line with the element overrides the rules define in an external CSS file as well as the rules defined in <style> element.

Example: style sheet rules along with the HTML elements using style attribute of those elements.

```
<! DOCTYPE html>
<html><head>
<title>HTML Inline CSS</title>
</head>
<body>
<p style="color:red;">This is red</p>
<p style="font-size:20px;">This is thick</p>
<p style="color:green;">This is green</p>
<p style="color:green;font-size:20px">This is thick and green</p>
</body>
</html>
```

This will produce the following result:

This is red

This is thick

This is green

This is thick and green

Example: the color and margin style sheet

```
<! DOCTYPE html>
<html><head>
<title> Inline Style Sheet</title>
</head>
<h1 style ="color:blue; margin-left:30px;">This is a heading.</h1>
<p> this is a paragraph.</p>
</body>
</html>
```

This will produce the following result:

This is a heading.

this is a paragraph.

Embedded Style Sheet

To make formatting of the HTML document using CSS easier, it is better to embed the entire style sheet into the document. This creates an embedded style sheet. To embed a CSS inside an HTML document, it can be defined the styles in the <head> section of the document. The style sheet section always starts by defining the style's types. To define an embedded style sheet, it always start with:

<style type="text/css">

The body of the style sheet goes within <style type="text/css"> and </style> and declares rules for the HTML element. For example: *p {font-size: 16pt}* this means that when we use the p element in our HTML document, the font size of the text enclosed within the <p> tag will be 16.

To add other attribute or properties to the p element, such as *p {font-size: 16pt;color:red;background-color:yellow}*
Example: here is a sample embedded style sheet:

```
<! DOCTYPE html>
<html><head>
<title> embedded style sheet example</title>
<style type="text/css">
body{background-color:yellow}
h1{font-size:50;color:blue}
p{font-size:14pt;color:red}
</style>
</head>
<body>
<h1>Embedded Style Sheet Example</h1>
<p> the entire HTML document can be formatted using embedded cascading style sheet, it
makes formatting a web page much easier </p>
</body>
</html>
```

This will produce the following result:

Embedded Style Sheet Example

the entire HTML document can be formatted using embedded cascading style sheet, it makes formatting a web page much easier

Example: here is the embedded style sheet example to format text of the HTML page

```
<! DOCTYPE html>
<html><head>
<title> embedded style sheet example</title>
<style type=”text/css”>
.red{
    color: red;
}
.thick{
     font-size:20px;
}
.green{
     color: green;
}
</style>
</head>
<body>
<p class=”read”>This is red</p>
<p class=”thick”>This is thick</p>
<p class=”green”> This is green</p>
<p class=”thick green”> This is thick and green</p>
</body>
</html>
```

The produced result will be:

This is red

This is thick

This is green

This is thick and green

Exercise:

1. Develop the drop down menu like shown below using cascading style sheet. When the curser is being over the menu the menu will be drop down and also when the curser is on the drop down items the color will be changed.(Note: Use inline and embedded style sheet rules)

LABORATORY VIII: External and Imported Style Sheet

Objectives:

- Write style sheet rules separated from the HTML script
- Define External style sheets to format the text and the web page
- Define the imported style sheet to make the web page attractive
- Define the text, background and other element of the web page using external and imported style sheet

External Style sheet

It is recommended to define a common style sheet in a separate file from the html file if the style sheet is needed to use to various pages. An external style sheet is simply a text file containing a list of CSS rules sets. The file is saved with a .css extension and saved to any directory that can be accessed by the web pages using it. Unlike embedded style sheets, the CSS rules sets do not have to be wrapped in the <style>...</style> tags. A cascading style sheet file will have extension as .css and it will be including in HTML files using <link> tag. The *link* element can be used to specify that a web page should use an external style sheet. The *link* element only requires a start tag <link> and is inserted in between the <head>…</head> tags of our web page document. It can be used as many times as we like.

The *link* element employs three important attributes:
rel : the value of this attribute is always *stylesheet type*
: the value of this attribute is always *text/css*.

href : the value of this attribute will be changed according to .css file to be referring to. This value can be any relative or absolute path.

Example1: Consider the following rules for the style sheet file called style.css

```
.red{
        color: red;
}
.thick{
        font-size:20px;
}
.green{
        color: green;
}
```

Here, three CSS rules are defined which will be applicable to three different classes defined for the HTML tags shown below. Now let's make use of the above external CSS file in our following HTML document.

Example 2: Consider the following rules for the style sheet file called main.css

```
<! DOCTYPE html>
<html><head>
<title>HTML external CSS example</title>
<link rel="stylesheet" type="text/css" href="/html/style.css">
</head>
<body>
<p class="red">This is red</p>
<p class="thick">This is thick</p>
<p class="green"> This is green</p>
<p class="thick green"> This is thick and green</p>
</body>
</html>
```

The produced result will be:?

```
H1 {
   font-family: “Times New Roman”;
   font-size: 36px;
   background: #999fff;
   color: maroon;
}

H2 {
   font-family: arial,verdana,sans-serif;
   font-size: 20px;
   background: #ffffff;
   color: black;
}

P {
   font-family: arial,verdana,sans-serif;
   font-size: 16px;
   background: #ffffff;
   color: navy;
}
```

Now let’s make use of the above external CSS file in our following HTML document.

```
<! DOCTYPE html>
<html><head><title>HTML external CSS example</title>
<link rel=”stylesheet” type=”text/css” href=”main.css”>
</head>
<body>
<h1>Internet Programming CSS</h1>
<h2>Start up Style Sheet Programming</h2>
<p> Internet programming is a course which is given to help students to develop
their own web page and manage web pages. This course covers the user side
programming to design and program user interacting interface and web pages for
their users. So that learners should attained and doing each of the programming
examples in this manual to proceed the server side programming of the web.</p>
<h2>External Style Sheet</h2>
In external style sheet, the rules of the style are defined separately from the HTML
document and the style sheet is saved with file name extension of .css. In the HTML
program, the style sheet is called using <link> tags which has three basic attributes
such as rel, type, href.</p>
<h2> Imported Style Sheet</h2>
<p>The imported style sheet is implemented in the next chapter. Hence before
implementing the imported style sheet, it is better to understand how the external
style sheet is defined and used to develop webpage and then focus on the next
session to implement the imported style sheet</p>
</body></html>
```

The produced output will be:

Internet Programming CSS

Start up Style Sheet Programming

Internet programming is a course which is given to help students to develop their own web page and manage web pages. This course covers the user side programming to design and program user interacting interface and web pages for their users. So that learners should attained and doing each of the programming examples in this manual to proceed the server side programming of the web.

External Style Sheet

In external style sheet, the rules of the style are defined separately from the HTML document and the style sheet is saved with file name extension of .css. In the HTML program, the style sheet is called using tags which has three basic attributes such as rel, type, href.

Imported Style Sheet

The imported style sheet is implemented in the next chapter. Hence before implementing the imported style sheet, it is better to understand how the external style sheet is defined and used to develop webpage and then focus on the next session to implement the imported style sheet

Imported Style sheet

A style sheet may be imported with CSS's @import statement. This statement may be used in a CSS file or inside the <style>…</style> tag.

Syntax:

```
<style type="text/css">
<!--
@import url("style1.css");
@import url("style2.css");
Body {background: yellow;color:black}
-->
</style>
```

Note that other CSS rules may still be included in the *style* element, but that all @import statements must occur at the start of the style sheet. Any rules specified in the style sheet itself override conflicting rules in the imported style sheets. For example, even if one of the imported style sheets contained *body {background : aqua}* , definition terms would still have a yellow background. The order in which the style sheets are imported is important in determining how they cascade. In the example above, if the style1.css imported style sheet specified that STRONG elements to be shown

in red and the style2.css style sheet specified that STRONG elements be shown in yellow, then the latter rule would win out, and STRONG elements would be in yellow. The imported style sheets are useful for purposes of modularity. For example if we have two and above CSS files that define different body of the page, each CSSs are used in the HTML using import element.

Example: the style.css file that define the headings H1 and H2

```
H1 {
  font-family: 'Times New Roman';
  font-size: 36px;
  background: #999fff;
  color: yellow;
}

H2 {
  font-family: arial,verdana,sans-serif;
  font-size: 20px;
  background: #ffffff;
  color: black;
}
```

The style1.css file that define the paragraph p and body color

```
P {
  font-family: arial,verdana,sans-serif;
  font-size: 16px;
  background: #ffffff;
  color: navy;
}

body {
        background: #ff0000
}
```

Exercise:

1. Develop the drop down menu like shown below using cascading style sheet. When the curser is being over the menu the menu will be drop down and also when the curser is on the drop down items the color will be changed.

PART THREE: JAVASCRIPT PROGRAMMING

LABORATORY IX: Basics and Program Control I of JavaScript

Objectives:

- Write javascript codes inside the HTML script in between head tag
- Write javascript codes inside the HTML script in between body tag
- Implement variables and operators in javascript programs
- Acquire skill to use conditional statements in javascript programs

Embedding JavaScript into HTML

JavaScript resides within an HTML document. Usually it is placed into an HTML document using <script> tag. It also can be added script to the head or the body section (or both) of an HTML document. Inline scripting is techniques that embed scripting instructions directly into certain HTML tags. The basic structure of an HTML file with JavaScript is as follows (JavaScript indicated in bold):

```
<html>
<head>
<title> Page Title</title>
<script language="JavaScript">
<!--
//JavaScript code goes here
//-->
</script>
</head>
<body>
HTML page text
<script language ="JavaScript">
<!--
//JavaScript goes here too
// -->
</script>
HTML page text
</body>
```

</html>

There are a flexible given to include JavaScript code anywhere in an HTML document. But there are following most preferred ways to include JavaScript in your HTML file.

- ✓ Script in <head>…</head> section
- ✓ Script in <body>…</body> section
- ✓ Script in <body>…</body> and <head>…</head> section>
- ✓ Script in and external file and then include in <head>…</head> section

Example: JavaScript code to write simple text

```
<html>
<head>
<title> First JavaScript page </title></head>
<body>
<h1> First JavaScript Page</h1>
<script type=”text/javascript”>
document.write(“<HR>”);
document.write(“Hello World Wide Web”); //comments goes here
document.write(“<HR>”);
</script>
</body><html>
```

The produced output will be:

The word **document.write** is a standard JavaScript command for writing output to a page. By entering the document.write command between the <script> and </script> tags, the browser will

recognize it as a JavaScript command and execute the code line. // is used for single line comment whereas /* *comment* ... */ is used for multiple line comment in JavaScript coding

Variables and operators

Variables in JavaScript should be declared before using them with the **var** keword. The scope of the variable is the region of the program in which it is defined . javascript variable will have only two scopes.

- ➔ Global variable –has global scope which is defined everywhere in the JavaScript code.
- ➔ Local variable – visible only within a function where it is defined. Function parameters are always local to that function.

JavaScript allows working with three primitive data type:

- ✓ Numbers e.g. 123, 120.50 etc
- ✓ String of text e.g. " javascript is my interest" etc
- ✓ Boolean e.g. true or false.

The arithmetic operators (+, - , *, / and %) are supported in JavaScript language. Suppose variable A holds 15 and variable B holds 5 then: A+B will give 20, A-B will give 10, A*B will give 65, A/B will give 3 and A%B will give 0

Example: JavaScript code that performs all the arithmetic operations and display their result

```
<!DOCTYPE html><html><head>
    <title>Arithmetic Operator Example</title>
</head>
<body>
    <script type="text/javascript">
        var num1 = 15;
        var num2 = 5;
        var sum = num1+num2;
        var diff = num1 - num2;
        var prod = num1 * num2;
        var quo = num1 / num2;
        var rem = num1 % num2;
        document.write("Sum = " + sum + "<br>");
        document.write("Difference = " + diff + "<br>");
        document.write("Product = " + prod + "<br>");
        document.write("Quotient = " + quo + "<br>");
        document.write("Remainder = " + rem + "<br>");
    </script>
</body>
</html>
```

To compare the value of two variables, comparison operators are used. These are == for equal, != not equal, > greater than, < less than, >= greater or equal and <= less than or equal.

The logical operators supported by javascript language are && called logical AND operator, | | called logical OR operator and ! called logical NOT operator

Conditional Statement

If statement

The **if** statement is the fundamental control statement that allows JavaScript to make decisions and execute statements conditionally.

Syntax:

if (expression){

Statement(s) to be executed if expression is true

}

if...else statement:

The **if...else** statement is the next form of control statement that allows JavaScript to execute statements in more controlled way.

Syntax:

if (expression){

Statement(s) to be executed if expression is true

}else{

Statement(s) to be executed if expression is false

}

if...else if... statement:

The **if...else if...** statement is the one level advance form of control statement that allows JavaScript to make correct decision out of several conditions.

Syntax:

if (expression 1){

Statement(s) to be executed if expression 1 is true

}else if (expression 2){

Statement(s) to be executed if expression 2 is true

}else if (expression 3){

Statement(s) to be executed if expression 3 is true

}else{ Statement(s) to be executed if no expression is true }

switch statement:

The basic syntax of the **switch** statement is to give an expression to evaluate and several different statements to execute based on the value of the expression. The interpreter checks each **case** against the value of the expression until a match is found. If nothing matches, a **default** condition will be used.

Syntax:

```
switch (expression)
{
case condition 1: statement(s)  break;
case condition 2: statement(s)  break;
…
case condition n: statement(s)  break;
default: statement(s)
}
```

Example: the if else JavaScript code

```
<html><head><title>conditional statement example</title></head><body>
<script type="text/javascript">
var d=new Date();
theday=d.getDay();
if(theday==0)
{document.write("First Monday");}
else if(theday==1)
{document.write("Second Teusday");}
else if(theday==2)
{document.write("third Wednesday");}
else if(theday==3)
{document.write("Happy therusday");}
else if(theday==4)
{document.write("Finally Friday");}
else if(theday==5)
{document.write("super saturday");}
else if(theday==6)
{document.write("Sleepy Sunday");}
else
{document.write("I'm looking forward to this weekend");}
</script></body><html>
```

Exercise: write the JavaScript code using switch case conditional statement to implement the above if else code.

LABORATORY X: Program Control II of JavaScript

Objectives:

- Acquire skill to use for loop in JavaScript programs
- Acquire skill to use break statements in a JavaScript programs
- Acquire skill to use nested loop statements in JavaScript programs
- Implement loops in JavaScript programs to format the web page

The for loop

The **for loop** is the most compact form of looping and includes the following three important parts:

- ✓ The loop initialization –counter starting value which is executed before the loop begins
- ✓ The test statement- the given condition is true or not is tested
- ✓ The iteration statement – increasing or decreasing the counter **Syntax:**

for (initialization; test condition; iteration statement){

Statement (s) to be executed if test condition is true

}

The while Loop

The most basic loop in JavaScript is the **while** loop which loops through a block of code as long as a specified condition is true

Syntax:

while (expression){

Statement(s) to be executed if expression is true

}

The do …while loop

The do…while loop is similar to the **while** loop except that the condition check happens at the end of the loop. This means that the loop will always be executed at least once, even if the condition is false.

Syntax:

do { Statement (s) to be executed; } while (expression);

The break Statement:

The **break** statement, which was briefly introduced with the ***switch*** statement, is used to exit a loop early, breaking out of the enclosing curly braces.

The continue Statement:

The **continue** statement tells the interpreter to immediately start the next iteration of the loop and skip remaining code block. When a **continue** statement is encountered, program flow will move to the loop check expression immediately and if condition remain true then it start next iteration otherwise control comes out of the loop.

Example 1: this example illustrates the use of a break statement with a while loop. Notice how the loop breaks out early once x reaches 5 and reaches to document. write (…) statement just below to closing curly brace.

```
<html>
<head><title>conditional statement example</title></head>
<body>
<script type="text/javascript">
var x = 1;
document.write("Entering the loop<br /> ");
while (x < 20)
{
  if (x == 10){
     break;  // breaks out of loop completely
  }
  x = x + 1;
  document.write( x + "<br />");
}
document.write("Exiting the loop!<br /> ");
</script>
</body></html>
```

The output of the above JavaScript code is:

```
Entering the loop
2
3
4
5
Exiting the loop!
```

Example 2: this example illustrates the use of a **continue** statement with a while loop. Notice how **continue** statement is used to skip printing when the index held in variable x reaches 5:

```
<html>
<head><title>conditional statement example</title></head>
<body>
<script type="text/javascript">
var x = 1;
document.write("Entering the loop<br /> ");
while (x < 10)
{
   x = x + 1;
  if (x == 5){
     continue;  // breaks out of loop completely
  }
    document.write( x + "<br />");
}
document.write("Exiting the loop!<br /> ");
</script></body></html>
```

This will produce the following result:

```
Entering the loop
2
3
4
6
7
8
9
10
Exiting the loop!
```

Example 3: This example illustrate the continue statement with nested for loop

```
<html>
<head><title>for loop</title></head>
<body>
    <script type="text/javascript">
        document.write("Enter the loop <br>");
        outerloop:
        for (var i = 0; i < 3; i++) {
            document.write("outerloop: " + i + "<br/>");
            for (var j = 0; j < 5; j++) {
                if (i == 3) {
                    continue outerloop;
                }
                document.write("<center>" + "Innerloop: " + j + "<br/></center>");
            } }
        document.write("Exiting the loop!<br/>");
    </script> </body> </html>
```

This will produce the following result.

```
Entering the loop!
Outerloop: 0
                    Innerloop: 0
                    Innerloop: 1
                    Innerloop: 2
Outerloop: 1
                    Innerloop: 0
                    Innerloop: 1
                    Innerloop: 2
Outerloop: 2
                    Innerloop: 0
                    Innerloop: 1
                    Innerloop: 2
Exiting the loop!
```

Exercise:

Develop a pattern like shown below using loop statement

```
*
**
***
****
*****
******
```

LABORATORY XI: Functions and Dialog Boxes

Objectives:

- Define and call functions in JavaScript programs.
- Define alert dialog box and implement in alerting warning message to the user
- Define confirmation dialog box to accept users option in the JavaScript program
- Define prompt dialog box to input something in the text box and respond accordingly

Function definition: before using a function, the function should be defined. The most common way to define a function in JavaScript is by using the function keyword, followed by a unique function name, a list of parameters may be empty, and a statement block surrounded by curly braces.

Syntax:

```
<script type="text/javascript"> function
functionname(parameter-list)
{
Statements
}
</script>
```

Example: A simple function that takes no parameters called sayHello is defined here:

```
<script type="text/javascript"> function
sayHello()  //function definition
{
Alert("Hello there");
}
</script>
```

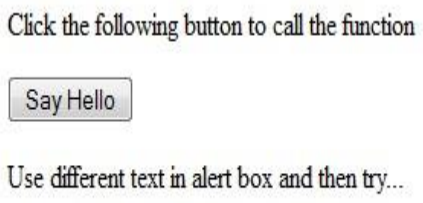
Function Calling: to invoke a function somewhere later in the script, we should simple need to write the name of that function as follows:

```
<script type="text/javascript">
sayHello(); //function calling
</script>
```

Example: this example illustrates function definition and function calling. When the button is clicked function calling is executed and then *hello there!* message will be displayed

```
<html>
<head>
<script type="text/javascript">
function sayHello() //function definition
{   alert("Hello there!");}
</script>
</head>
<body>
<p>Click the following button to call the function</p>
<form>
<input type="button" onclick="sayHello()" value="Say
Hello"> //sayhello() is function calling
</form>
<p>Use different text in alert box and then try...</p>
</body>
</html>
```

The produced result will be:

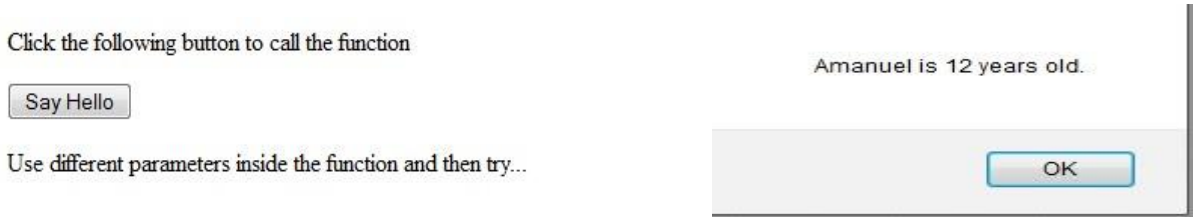


When the button is clicked, the result displayed in the alert box will be the following

Example 2: this example illustrates the function definition and function calling with parameters as given bellow the JavaScript code and the output of the program

```
<html>
<head>
<script type="text/javascript">
function sayHello(name, age)
{
  alert( name + " is " + age + " years old.");
}
</script>
</head>
<body>
<p>Click the following button to call the function</p>
<form>
<input type="button" onclick="sayHello('Amanuel ', 12)" value="Say
Hello">
</form>
<p>Use different parameters inside the function and then try...</p>
</body>
```

The output produced will be: when the button is clicked the output will be:



Alert Dialog Box: an alert dialog box is mostly used to give a warning message to the users. Like if one input field requires to enter some text but user does not enter that field then as a part of validation you can alert box to give warning message as follows:

Example: the following JavaScript code shows the alert dialog box message.

```
<!DOCTYPE html>
<html>
<head>
<head>
<script type="text/JavaScript">
alert("Warning Message");
</script>
</head>
</html>
```

The following result will be produced:

Confirmation Dialog Box: A confirmation dialog box is mostly used to take user's consent on any option. It displays a dialog box with two buttons: **Ok** and **Cancel.** We can use confirmation dialog box as follows:

Example: this example illustrates the combination of confirmation and alert message. When the user clicked the *try it now* button the confirmation dialog box will be displayed and then on the confirmation dialog box either of the two buttons, the message will be displayed on the alert **Box:** dialog box.

```
<html>
<head><title>hhhh</title>
</head>
<body>
<form name=myform>
<input type=button value="Try it now"
onClick="if(confirm('Format the hard disk?'))
alert('You are very brave!');
else alert('A wise decision!')">
</form>
</body>
</html>
```

The following result will be produced

Prompt Dialog

the prompt dialog box can be implemented as follows

The following result will be produced

Enter your name

then

```
<html>
<head>
<script type="text/javascript">
var retVal = prompt("Enter your name : ", "your name here");
 alert("You have entered : " + retVal );
 </script>
</head>
</html>
```

Exercise:

1. write the JavaScript code that accepts the name and grade of the user to display the following message in alert dialog box *"Congratulation name of user !!! you score grade"* using function definition and function calling.

2. Write the JavaScript code that accept your name, your department and the Name of course and displays them in order like

   Hello Solomon from Computer Science Department you take Internet programming course

LABORATORY XII: Form Validation

Objectives:

- write code to validate whether data is/are entered into the form or not
- write code to validate the entered data is/are valid or not
- write code to validate input forms

Form validation used to occur at the server, after the client had entered all necessary data and then pressed the Submit button. If some of the data that had been entered by the client had been in the wrong form or was simply missing, the server would have to send all the data back to the client and request that the form be resubmitted with correct information. This was really a lengthy process and over burdening server.

JavaScript provides a way to validate form's data on the client's computer before sending it to the web server. Form validation generally performs two functions.

- ✓ **Basic Validation -** First of all, the form must be checked to make sure data was entered into each form field that required it. This would need just loop through each field in the form and check for data.
- ✓ **Data Format Validation -** Secondly, the data that is entered must be checked for correct form and value. This would need to put more logic to test correctness of data.

The code shown below in the example produces the output shown in the figure below contains Name, email and country with submit form

Example: understand the process of validation. Here is the simple form to proceed:

```
<html>
<head>
<title>Form Validation</title>
 <script type="text/javascript">
// Form  validation code will come here.
 </script> </head> <body>
 <form action="/cgi-bin/test.cgi" name="myForm"
onsubmit="return(validate());">
 <table cellspacing="2" cellpadding="2" border="1">
<tr>
<td align="right">Name</td>
 <td><input type="text" name="Name" /></td> </tr>
<tr> <td align="right">EMail</td> <td><input type="text"
name="EMail" /></td> </tr>
  <tr> <td align="right">Country</td> <td> <select
name="Country">
 <option value="-1" selected>[choose yours]</option>
<option value="1">Ethiopia</option> <option
value="2">Sudan</option>
 <option value="3">Kenya</option>
</select> </td> </tr>
<tr> <td align="right"></td>
 <td><input type="submit" value="Submit" /></td>
</tr>
</table>
</form>
 </body>
</html>
```

Basic Form Validation:

First we will show how to do a basic form validation. In the above form we are calling **validate()** function to validate data when **onsubmit** event is occurring. Following is the implementation of this validate () function:

```
<script type="text/javascript">
function validate()
{
if( document.myForm.Name.value == "" )
{
alert( "Please provide your name!" );
document.myForm.Name.focus() ;
return false;
}
if( document.myForm.EMail.value == "" )
{
alert( "Please provide your Email!" );
document.myForm.EMail.focus() ;
return false;
}
if( document.myForm.Country.value == "-1" )
{
alert( "Please provide your country!" );
return false;
}
return( true );
}
</script>
```

Data Format Validation:

Now we will see how we can validate our entered form data before submitting it to the web server. This example shows how to validate an entered email address which means email address must contain at least an @ sign and a dot (.). Also, the @ must not be the first character of the email address, and the last dot must at least be one character after the @ sign:

```
<script type="text/javascript">
function validateEmail()
{
var emailID = document.myForm.EMail.value;
atpos = emailID.indexOf("@");
dotpos = emailID.lastIndexOf(".");
if (atpos < 1 || ( dotpos - atpos < 2 ))
{
alert("Please enter correct email ID")
document.myForm.EMail.focus() ;
return false
}
return( true );
}
</script>
```

The following example illustrates the validation of password data and confirms password date. If the values are not equal the message *"your password do not match"* will be displayed at the result text box otherwise it will be ok.

```
<html>
<head>
<title> User Account</title>
<script type ="text/JavaScript">
function CheckPassword()
{
  var Password, ConfPassword, Result;
  Password = document.frmRegistration.txtPassword.value;
  ConfPassword = document.frmRegistration.txtConfirmPass.value;
  Result = document.frmRegistration.txtResult;
  if(Password != ConfPassword)
    Result.value = "Your Passwords Do Not Match";
}
</script></head><body>
<h1>User password Registration</h1>
<form name="frmRegistration">
 <table border="0" width="403">
<tr> <td width="99">Username:</td><td width="290"> <input type="text"
name="txtUsername" size="15"></td></tr>
 <tr> <td width="99">Password:</td><td width="290"> <input type="password"
name="txtPassword" size="15"></td></tr>
 <tr> <td width="99">Confirm:</td><td width="290">  <input type="password"
name="txtConfirmPass" size="15"></td> </tr>
  <tr><td width="99"> Result:</td> <td width="290"> <input type="text"
name="txtResult" size="40"> </td></tr>
    </table><input type="button" value="Send It" onClick="CheckPassword()">
</form></body></html>
```

The following result will be produced

Exercise:

Construct the user registration form that registers first name, second name, country, and email using HTML. Validate that first name, second name and country should only accept alphabet whereas email contains @ and dot (.)